

Bank Transaction Fraud Dashboard

Fraud Overview

Total Transactions

250,000

Fraud Transactions

13,813

Fraud Rate

5.53%

Fraud Amount

\$3,174,491.35

Transactions Amount

\$51,174,465.30

Payment Method

Device Type

Key Findings:

1. Total Volume & Rate: A total of 250,000 transactions were analyzed during the period, with 13,813 flagged as fraudulent (representing a 5.53% fraud rate).

2. Financial Impact: Total losses due to fraud reached \$3.17M, indicating a significant financial impact on operations.

3. High-Risk Categories: Fraud cases are heavily concentrated in specific merchant categories, led by Crypto Exchange, Jewelry, and ATM Withdrawals.

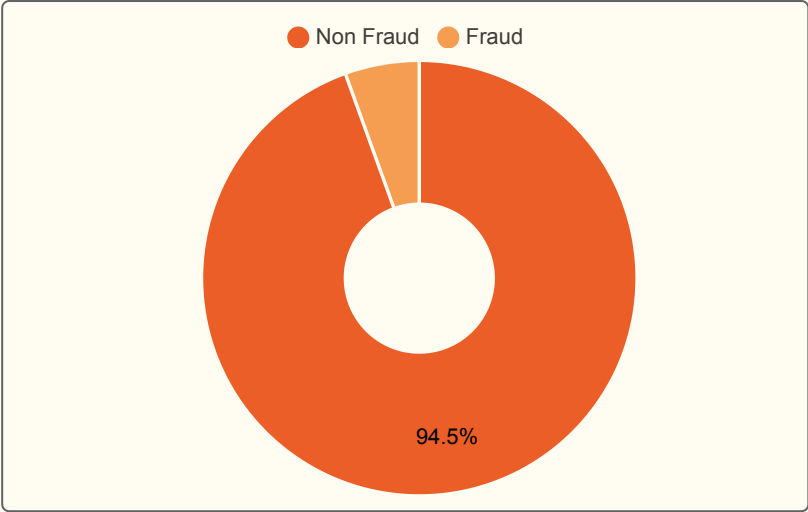
4. Geographic Concentration: Fraud activity is highly localized in key countries and cities, making these regions primary targets for enhanced monitoring.

Recommendation:

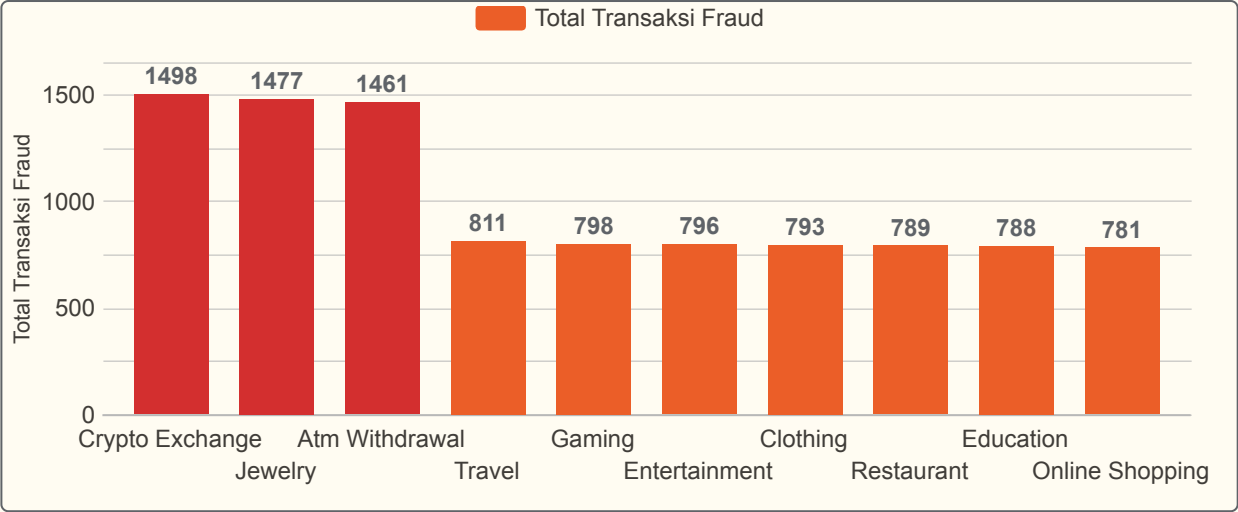
1. Strengthen Merchant Controls: Reinforce real-time fraud detection systems for high-risk merchant categories (Crypto Exchanges, Jewelry, and ATMs).

2. Targeted Geographic Monitoring: Allocate localized monitoring resources and risk rules to regions with the highest concentration of fraud

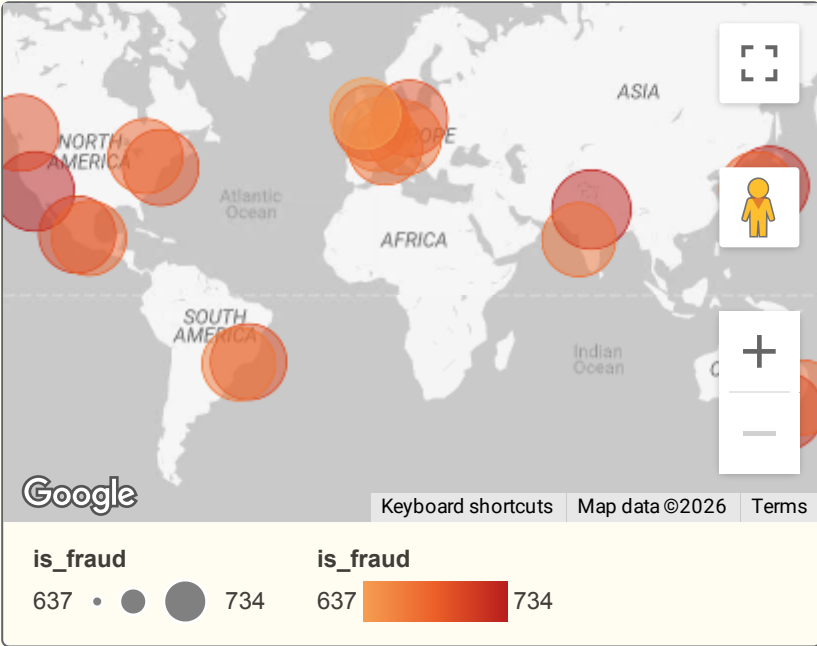
Fraud vs Non Fraud



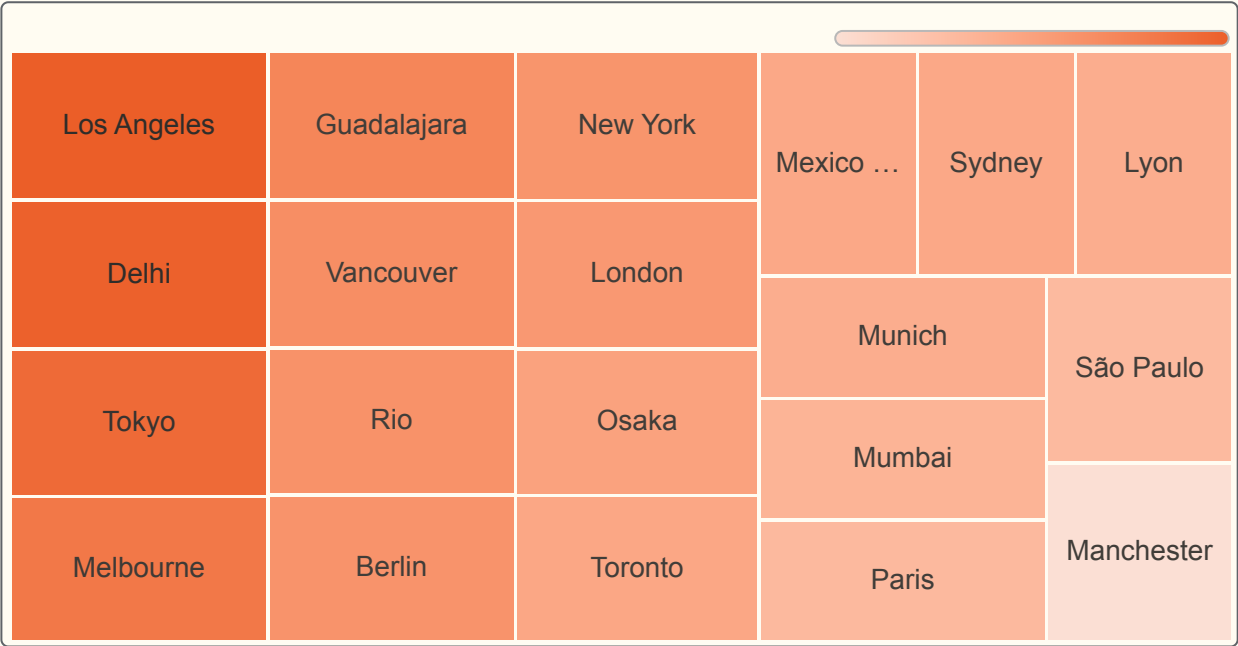
Fraud by Merchant Category



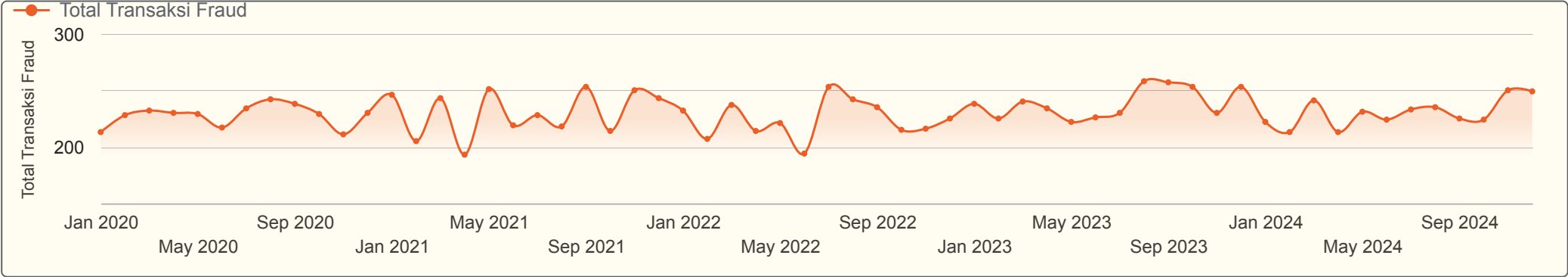
Geographic Fraud Distribution



Fraud by City



Fraud Trend



Bank Transaction Fraud Dashboard

Fraud Transactions Pattern Analysis



Transaction Date

Merchant Category

Key Findings:

1. Payment Method Vulnerability: Credit Cards represent the highest risk payment channel, driving the majority of fraudulent transaction volumes.

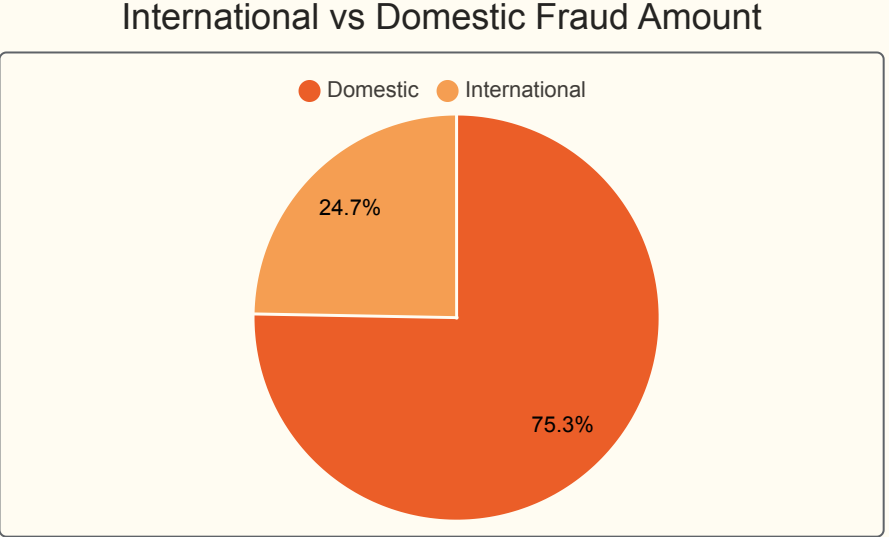
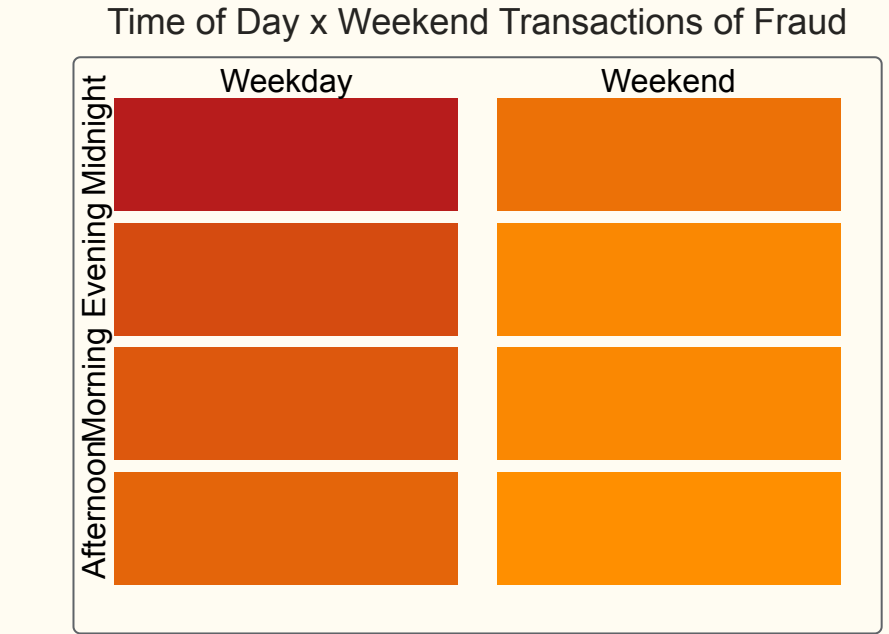
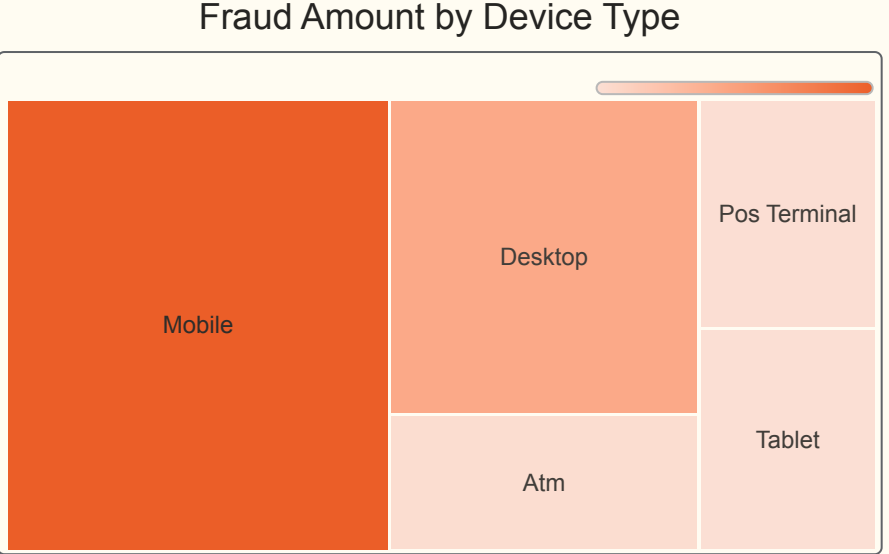
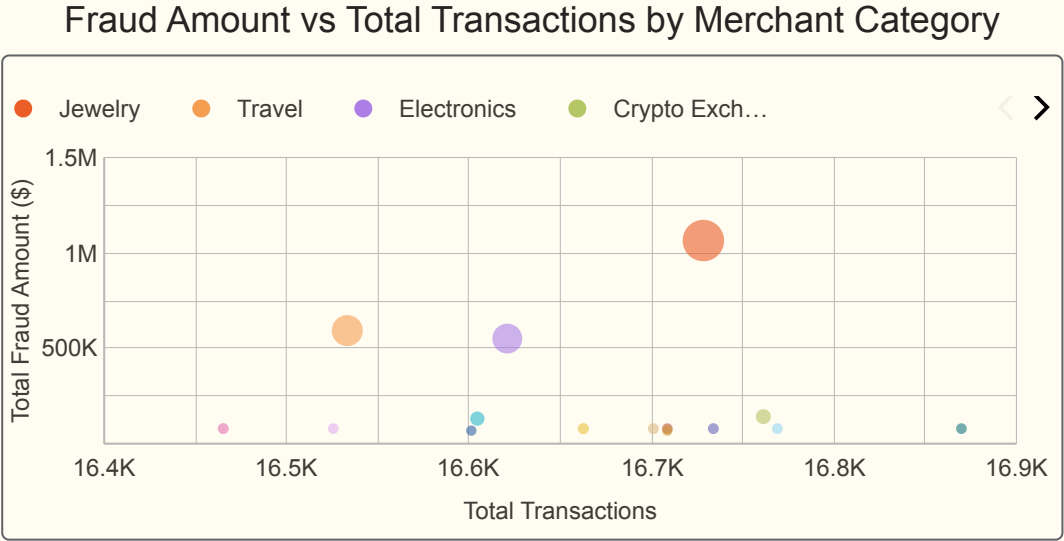
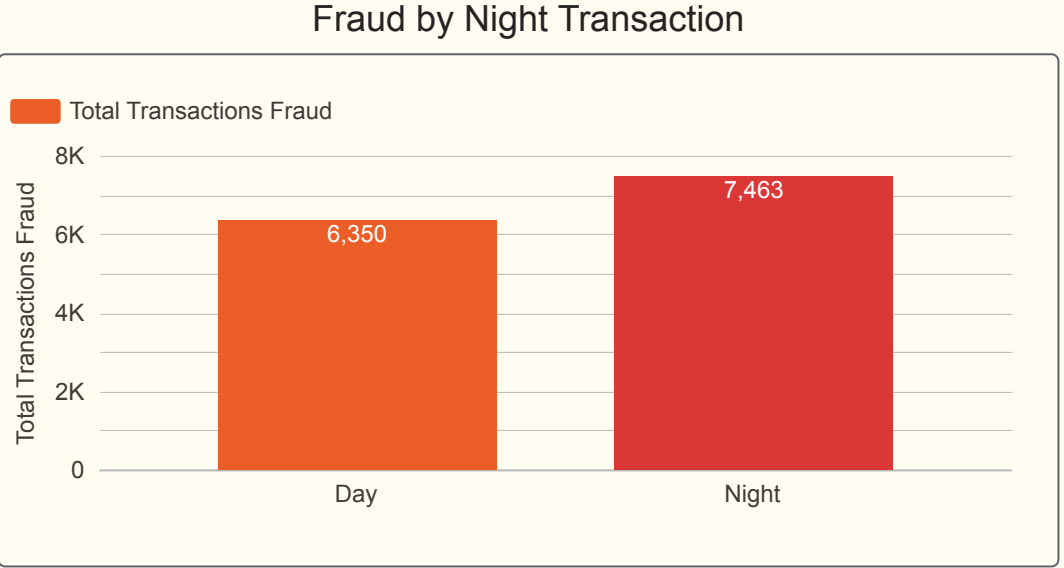
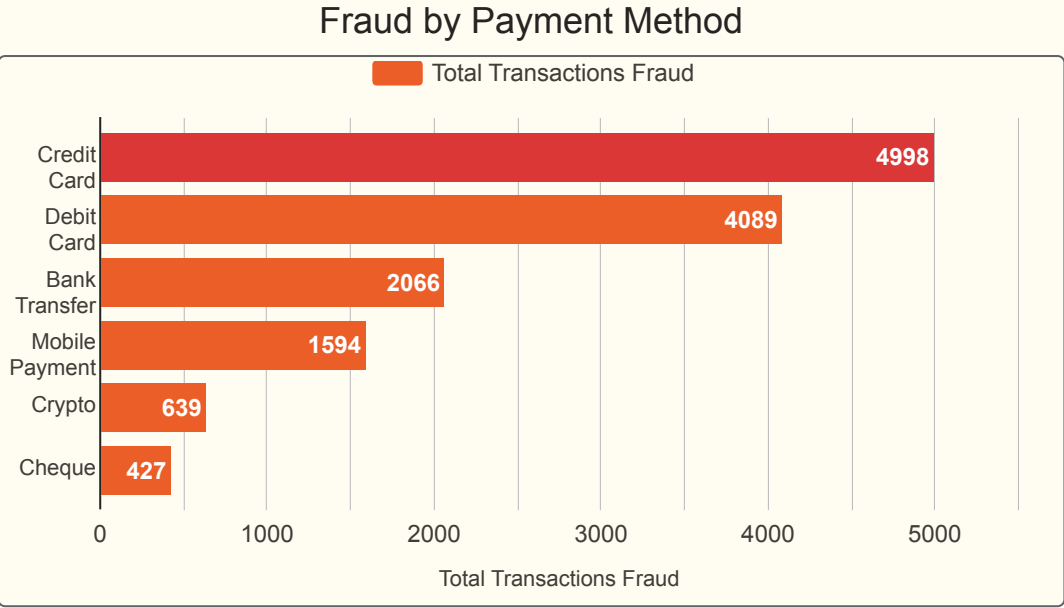
2. Device Preference: Transactions originating from Mobile and Desktop devices account for the vast majority of fraud cases compared to physical POS terminals.

3. Cross-Border Risk: Domestic transactions contribute to the bulk of fraud volume, though international transactions present a higher average exposure per transaction.

Recommendation:

1. Enhance Online Authentication: Implement stricter Step-Up Authentication specifically for Mobile and Desktop transactions.

2. Credit Card Fraud Rules: Tighten velocity rules and transaction limits on Credit Card transactions during off-peak hours.



Bank Transaction Fraud Dashboard

Customer Risk & Detection Analysis

Customer Age

41.78

Avg Failed Attempts

0.82

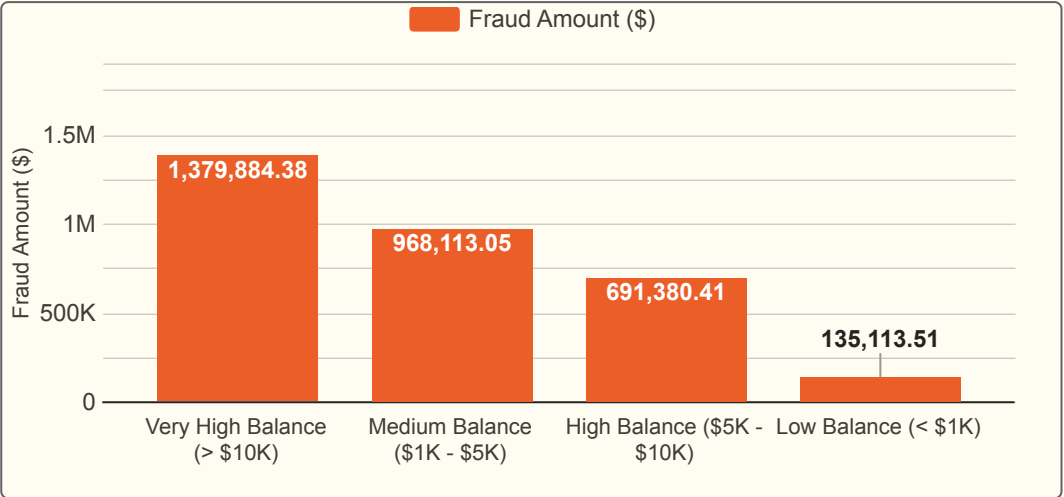
Distance From Home (Km)

20.04

High Risk Account Age

High Risk Account Age / Fraud Transactions	
Account Age Group	Fraud
> 3 Years	7,558
1 - 3 Years	3,919
< 1 Year (New Account)	2,336

Fraud Amount by Account Balance



Account Balance Tier Fraud Transacti...



Type to search



Very High Balance (> \$10K) 6K



Medium Balance (\$1K - \$5K) 4.1K



High Balance (\$5K - \$10K) 3.1K

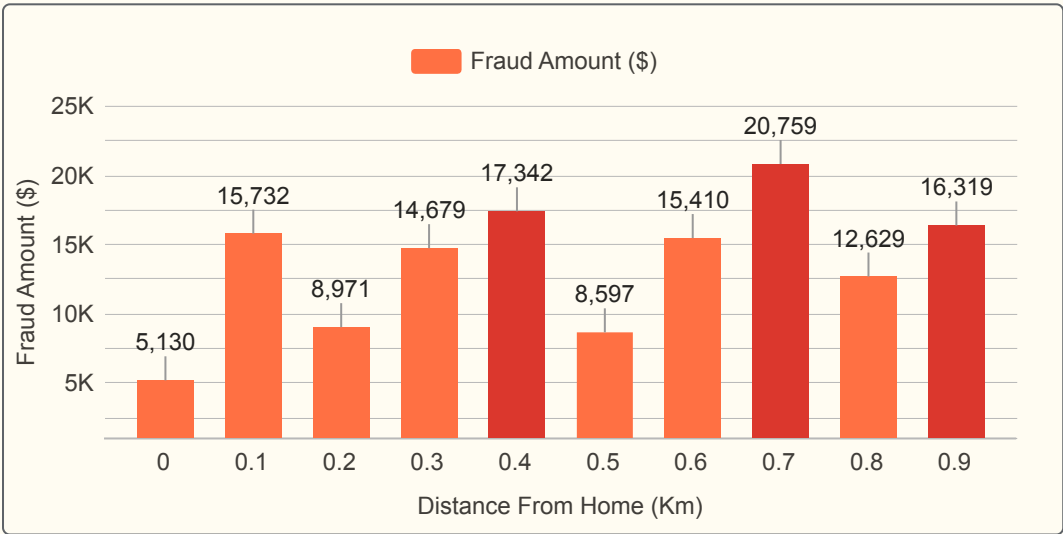


Low Balance (< \$1K) 565

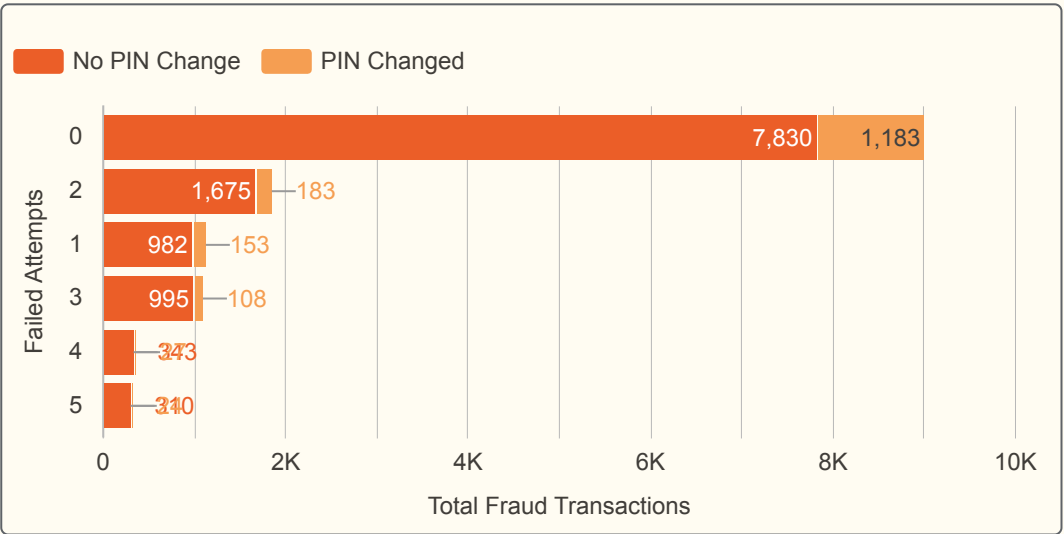
Credit Score Tier



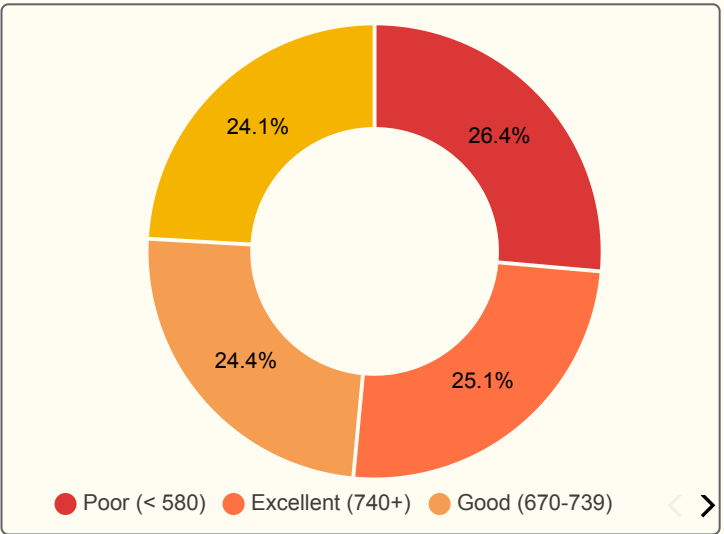
Fraud by Distance From Home



Failed Attempts vs Pin Changed Recently



Fraud by Credit Score



Key Findings:

- High Risk Credit Score: Customers with Poor and Fair credit scores account for over 50% of total fraud exposure.
- Authentication Anomalies: Fraud instances spike significantly on accounts experiencing prior failed login/PIN attempts and recent PIN changes.
- Geographic Deviation: High fraud amounts occur at an average distance of ~20 km from registered home locations.

Recommendation:

- Behavioral Step-Up Auth: Trigger mandatory OTP/Biometric verification when a transaction occurs following failed PIN attempts or a recent PIN update.
- Targeted Risk Monitoring: Apply stricter transaction velocity limits to accounts classified under Poor/Fair credit score tiers.